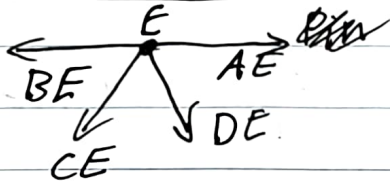


Calculating shear stress needed for the pins

Pin E has the most connections (4) & therefore the highest stress



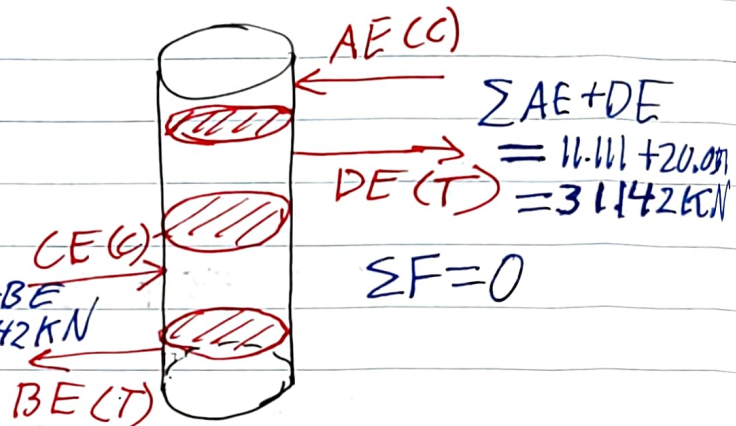
$$AE = -11.111 \text{ kN (C)}$$

$$BE = 11.111 \text{ kN (T)}$$

$$CE = -20.031 \text{ kN (C)}$$

$$DE = 20.031 \text{ kN (T)}$$

$$\Sigma CE + BE = 31.142 \text{ kN}$$



3A

$$\tau_{max} = 170 \text{ ksi} = 1172.11 \text{ MPa}$$

$$V = 31.142 \text{ kN}$$

$$\tau = \frac{V}{A_{min}}$$

$$\tau = \frac{V}{\pi r^2} = \frac{V}{\frac{\pi}{4} d^2}$$

$$d^2 = \frac{V}{\tau \cdot \frac{\pi}{4}} = \sqrt{\frac{V}{\tau \cdot \frac{\pi}{4}}} = \sqrt{\frac{31.142 \cdot 10^3 \text{ N}}{1172.11 \frac{\text{N}}{\text{mm}^2} \cdot \frac{\pi}{4}}}$$

$$d = 0.184 \text{ mm W/o safety}$$

Not correct

shear yield strength

$$\tau = \frac{V}{A_{min}} \quad | \quad \tau_y = 1172.11 \text{ MPa}$$

$$\frac{\tau_y}{SF} = \frac{V}{3A_{min}} \Rightarrow \frac{V \cdot SF}{3\tau_y} = A_{min} \quad \left| \begin{array}{l} V = 31.142 \cdot 10^3 \text{ N} \\ SF = 4 \end{array} \right.$$

$$A_{min} = \frac{(31.142 \cdot 10^3 \text{ N})(4)}{3(1172.11 \frac{\text{N}}{\text{mm}^2})} = \boxed{106.28 \text{ mm}^2} = \boxed{35.426 \text{ mm}^2}$$

$$A_{min} = \pi r^2$$

$$\frac{35.426 \text{ mm}^2}{106.28 \text{ mm}^2} = \pi r^2$$

$$r = \sqrt{\frac{35.426}{\pi}} = \boxed{3.358 \text{ mm}} \quad \text{Rounding up: } \boxed{3.36 \text{ mm}}$$

$$d = 2r = 2(3.36) = \boxed{6.72 \text{ mm}}$$

weight of pins $\rho = 0.278 \text{ lbs/in}^3 = 7695.0135 \text{ Kg/m}^3$

minimum Length needed for pin E

diameter of Beams

$$20 \text{ mm} \cdot 4 = 80 \text{ mm}$$

4 beams at pin E

$$V = \pi r^2 h = \pi (3.36 \text{ mm})^2 (80 \text{ mm}) = \boxed{2837.386 \text{ mm}^3}$$

$$\text{weight} = \rho \cdot V = (7695.0135 \frac{\text{Kg}}{\text{m}^3}) (2837.386 \cdot 10^{-9} \text{ m}^3) = \boxed{0.0218 \text{ Kg}} \text{ per pin}$$